



CERTIFICATE OF TESTS

Page 11

Printed: 03/24/2025

Heat: 24160832
Material: 5L-X52 4-1/2X.337 (15.00#)BR,DRL,PEB (11000102)
Vendor: Big River Steel - Osceola AR US (180431)

Purchase Order: 26708947
Sales Order: 1001001265
MFG Date: 01/30/2025

CHEMICAL PROPERTIES

	C	MN	P	S	NI	CR	CU	MO	SI	TI	V	CE
Ladle	0.0500	0.5600	0.0090	0.0020	0.0400	0.0400	0.0900	0.0140	0.1300	0.0010	0.0030	0.0917
	Al: 0.0370		Sn: 0.0053		N: 0.0078		B: 0.0002		Ca: 0.0028		Nb: 0.0150	
Chem	0.0658	0.5587	0.0059	0.0009	0.0403	0.0357	0.0786	0.0091	0.1216	0.0008	0.0028	0.1049
	Al: 0.0329		Sn: 0.0042		N: 0.0203		B: 0.0000		Ca: 0.0024		Nb: 0.0138	
Chem	0.0647	0.5762	0.0060	0.0005	0.0411	0.0354	0.0801	0.0093	0.1399	0.0007	0.0030	0.1055
	Al: 0.0334		Sn: 0.0044		N: 0.0157		B: 0.0000		Ca: 0.0026		Nb: 0.0141	

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 68.60 TS (ksi): 72.50 EL (%) : 26.00 HRB: 84.00 Y/T(%): 0.95

CHARPY TEST RESULTS (BODY)

Orientation : TRANSVERSE	Specimen(S)	Energy (ft-lb)	%Shear
Spec. Size (in) : .394"X.197"X2.165"	1	79.00	100.00
Test Temp (°F) : -20	2	83.00	100.00
Location : 3T90	3	80.00	100.00
	Average	80.67	100.00

Acceptance Criteria Min. (ft-lb) : 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-24,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Big River Steel - Osceola AR US
COIL MANUFACTURER: Big River Steel - Osceola AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.

Quality Control Dept